The Convergence of Minds and Bodies: A Systematic Review of Generative AI Agents in Extended Reality for Education

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***Abstract*—The integration of Generative Artificial Intelligence (GenAI) into Extended Reality (XR) ecosystems marks a transformative shift from deterministic, pre-scripted simulations to dynamic, intelligent embodiment. This survey systematically investigates how Large Language Models (LLMs) and multimodal reasoning agents act as a cognitive layer to reshape pedagogical mechanisms within immersive spaces. By synthesizing 73 high-impact articles, we establish a functional taxonomy of embodied pedagogical agents—categorized into Tutors, Peers, Role-Players, and Co-creators—and map their deployment across diverse educational domains such as STEM, humanities, and vocational training. Our analysis reveals a parabolic growth in "Generative XR" research since late 2023, signifying a transition from simple technology integration to the reconfiguration of spatial affordances. We deconstruct how specific XR modalities dictate agent design: Virtual Reality (VR) dominates as a "generative sandbox" for total environmental control (64%); Augmented Reality (AR) provides contextual grounding and ontological security (46%); and Mixed Reality (MR) facilitates embodied co-presence (10%). Furthermore, we examine the impact of these systems on cognitive, skill, and affective outcomes, noting significant enhancements in knowledge transfer, reduced cognitive load via adaptive scaffolding, and increased social self-efficacy for neurodivergent learners. Despite this potential, the field faces critical technical bottlenecks, including multimodal latency, AI hallucinations, and the "prompting barrier" for non-technical users. We conclude by proposing design guidelines for modality selection and agent role orchestration, advocating for longitudinal evidence to validate the long-term pedagogical efficacy of generative agents in immersive learning environments.**

***Index Terms*—Large Language Models, Spatial Computing, Intelligent Tutoring Systems, Human-Computer Interaction, Systematic Review**

I. Introduction

THE integration of Extended Reality (*XR*) into educational ecosystems has long been driven by the pursuit of *situated learning*, creating a pedagogical framework where learning is most effective when it occurs within the context in which it will be applied. *XR* technologies, comprising Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), offer the unique affordance of *spatial presence* by allowing learners to inhabit high-fidelity simulations of physical environments. However, despite decades of hardware advancement, the pedagogical efficacy of *XR* has been consistently bottlenecked by the content rigidity problem. Traditional *XR* learning environments are predominantly deterministic. They function as pre-scripted state machines where learner agency is confined to finite decision trees, and Non-Player Characters (NPCs) operate as static information kiosks rather than dynamic mentors.

The emergence of Generative Artificial Intelligence (*GenAI*), specifically Large Language Models (LLMs) and multimodal reasoning agents, introduces a transformative cognitive layer to the spatial advantages of *XR*. This convergence facilitates the transition from *static simulation* to *intelligent embodiment*. Unlike their rule-based predecessors, *GenAI-driven* agents possess the capacity for open-ended natural language understanding, context-aware content generation, and real-time pedagogical adaptation.

This paradigm shift redefines the "Iron Triangle" of educational technology, consisting of Scale, Quality, and Cost. By deploying autonomous agents capable of passing the Turing test in specific instructional domains, educators can potentially address Bloom's "2 Sigma Problem" [1] by providing personalized, one-on-one tutoring at mass scale within immersive environments. This survey systematically investigates this transition and analyzes how Embodied Pedagogical Agents (EPAs) leverage generative capabilities to reshape the mechanisms of instruction, collaboration, and cognitive scaffolding in *XR*.

A. The Paradigm Shift: From Deterministic Scenarios to Probabilistic Interaction

1) The Authoring Bottleneck and the Deterministic Trap: In the pre-generative era, the development of *Intelligent Tutoring Systems (ITS)* within *XR* was constrained by the "authoring bottleneck." Creating a responsive virtual tutor required instructional designers to manually anticipate every potential learner query and hard-code specific responses. This resulted in brittle systems where learners often hit dead ends in dialog, which broke the sense of immersion and induced *negative presence*.

2) The Generative Solution: Probabilistic Scaffolding: *GenAI* fundamentally circumvents this bottleneck by shifting the interaction model from deterministic retrieval to probabilistic generation. Modern agents function as reasoning engines that can construct knowledge incrementally. This allows for *Adaptive Scaffolding*, where the agent dynamically adjusts the

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complexity of its guidance based on the immediate performance and cognitive load of the learner, mirroring the behavior of a human expert.

B. Taxonomy of Intelligent Agent Roles

To structure our analysis, we classify *GenAI* agents not by their visual fidelity but by their *Pedagogical Function*. We identify four primary archetypes that distinctively influence the learning process: **(1) The Omniscient Tutor:** An authoritative agent that provides direct instruction, formative feedback, and assessment. This role optimizes *Cognitive Load Management* by guiding the learner's attention to germane information. **(2) The Collaborative Peer:** A "learning companion" designed to reduce the authority gradient. By simulating a peer who is also "learning," this agent fosters *Social Presence* and encourages critical thinking through collaborative problem-solving and error modeling. **(3) The Situated Role-Player:** Agents that simulate specific personas, such as a patient in medical triage, a historical figure in cultural heritage, or an upset customer in soft-skills training. These agents provide *Contextual Fidelity* allowing learners to practice high-stakes interpersonal skills in risk-free environments. **(4) The Co-Creator:** A tool-like agent that assists learners in generative design tasks, such as 3D modeling via voice commands, acting as a "force multiplier" for the creative intent of the learner.

C. Background and Related Work

1) Theoretical Foundations: Reimagining the Community of Inquiry in Spatial Computing: The theoretical framework of this survey represents a convergence of the *Community of Inquiry (CoI)* model and *Presence Theory*, adapted specifically for the era of *Generative XR*. While the traditional *CoI* framework posits that deep learning emerges from the effective intersection of *Social*, *Cognitive*, and *Teaching Presence*, the introduction of immersive technologies necessitates a re-evaluation of these constructs through the lens of spatial embodiment. Recent systematic reviews [2] have expanded this taxonomy to identify distinct presence modalities, specifically *Spatial, Social, Co-presence, Self, and Cognitive Presence*, identifying them as critical determinants that directly modulate learning outcomes and user engagement in virtual environments.

a) The Foundation of Situated Cognition: Spatial and Cognitive Presence: In the context of immersive learning, Spatial Presence serves as the sensory foundation for situated cognition. It creates the "illusion of non-mediation," allowing learners to perceive themselves as being physically located within the instructional environment rather than merely observing it. This sense of "being there" is not merely esthetic but functional. Research utilizing high-fidelity rendering and real-time ray tracing demonstrates that enhanced spatial presence enables learners to form accurate spatial mental models, which are prerequisites for complex simulation tasks [3]. Crucially, this spatial immersion acts as a catalyst for *Cognitive Presence*, which encompasses the extent to which learners are able to construct meaning through sustained reflection and discourse. Empirical evidence suggests a strong synergistic relationship between these two modalities. Systematic analyzes reveal that the intersection of Spatial and Cognitive Presence is particularly impactful for higher-order tasks such as decision-making and problem-solving simulations [2]. In this synergy, the spatial environment provides the contextual scaffolding, while the *GenAI* agent facilitates the exploration and integration phases of inquiry through adaptive questioning and Socratic dialog, transforming passive immersion into active intellectual engagement.

b) The Mechanics of Embodiment: Social and Co-Presence: The integration of *GenAI* further revolutionizes *Social Presence*, shifting it from the textual exchanges of asynchronous learning to a visceral sense of "being with" another intelligent entity. Within *XR*, social presence is inextricably

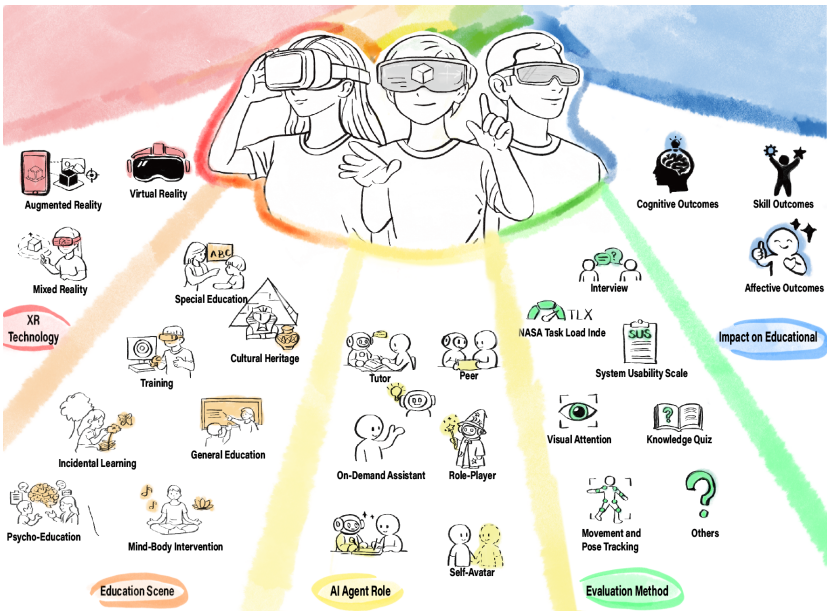


Fig. 1. Visual summary of the systematic review.

linked with *Co-presence*, defined as the sensory accessibility of the other. Literature indicates that this form of presence is significantly amplified by high-fidelity anthropomorphic avatars that accurately replicate human non-verbal behaviors, such as facial expressions and body movements [4]. Unlike traditional rule-based agents, *GenAI-driven* agents can dynamically generate context-aware proxemics, gaze, and gestures. This capability allows them to facilitate emotional communication and reduce the ambiguity often found in human-computer interaction. The combination of *Social Presence* and *Co-presence* has been identified as the most extensive factor in successful collaborative tasks, suggesting that *GenAI* agents effectively function by establishing an "emotional resonance" that maintains group cohesion and learner motivation.

2) Critical Gap Analysis: The Convergence of Minds and Bodies: Despite the rapid proliferation of literature in both *Extended Reality and AI*, a significant disconnect persists between these two research silos. Existing surveys typically exhibit one of two critical deficiencies that this study aims to address.

a) Type I Gap: The "Mindless Body" (XR-Centric Reviews): The first category of literature, exemplified by surveys such as Radianti et al. [5] and Checa and Bustillo [6], provides exhaustive analyzes of VR and AR hardware, usability, and rendering techniques. However, these studies predominantly discuss the "intelligence" of the system in terms of static decision trees or "Wizard of Oz" control mechanisms. In these frameworks, the virtual agents possess a high-fidelity "body" but lack a responsive "mind." They fail to account for the emergent behaviors, adaptability, and open-ended conversational capabilities of modern Large Language Models. Consequently, these reviews overlook the potential for agents to handle unscripted pedagogical interventions, limiting their applicability to deterministic scenarios where learner agency is constrained.

b) Type II Gap: The "Disembodied Mind" (GenAI-Centric Reviews): The second category, represented by recent works on LLMs in education [7], [8], analyzes the cognitive capabilities of models like *ChatGPT* but remains largely confined to text-based or 2D desktop interfaces. These reviews neglect the crucial dimension of Embodiment. They do not explore how an agent's physical occupation of 3D space, spatial audio, and joint attention influence learning dynamics. In a text-only interface, an AI cannot demonstrate a physical procedure, guide a learner's gaze to a specific object in 3D space, or leverage proxemics to build rapport. While emerging technical reviews [9] have begun to map *LLMs* to *XR*, their analysis is primarily oriented towards *HCI* metrics such as latency and rendering optimization rather than pedagogical efficacy. This survey addresses this gap by systematically deconstructing the *Pedagogical Mechanics* of *Generative Agents*, positioning them as the bridge that unites the "Mind" of *GenAI* with the "Body" of *XR*.

D. Objective and Research Questions

This study bridges the gap between the theoretical capabilities of *Generative AI* and its pedagogical implementation in *XR*. Moving beyond technical architectures, we examine how the convergence of probabilistic intelligence and spatial immersion creates novel affordances for high-order learning. The analysis is driven by three core research questions:

RQ1 (Taxonomy & Landscape): What pedagogical roles do *GenAI* agents assume in *XR*, and how are they distributed across domains (e.g., *STEM*, Humanities, Vocational Training)? Moving beyond visual fidelity, we establish a taxonomy based on functional pedagogy—categorizing agents into *Tutors*, *Peers*, *Role-Players*, and *Co-Creators*—to identify innovation drivers and application gaps.

RQ2 (Design & Interaction): What specific design affordances (e.g., multimodal interaction, non-verbal behavior, embodiment fidelity) enable pedagogical efficacy? We map technical implementation strategies to the enhancement of "feeling real" (Social Presence) and "thinking deeply" (Cognitive Presence).

RQ3 (Mechanism & Efficacy): Through what mechanisms do these agents influence learning outcomes? Moving beyond satisfaction metrics, we analyze how generative spatial immersion impacts *Cognitive Load*, *Self-Efficacy*, and *Knowledge Retention*, focusing on how adaptive scaffolding drives skill acquisition and transfer.

II. METHODOLOGY

Following the PRISMA guidelines, we queried the IEEE Xplore and ACM Digital Library databases using a Boolean strategy intersecting three core clusters: "Generative AI/Agents," "XR Environments," and "Educational Contexts."

To maintain empirical rigor, studies were included if they met three simultaneous criteria: (1) integration of both a generative AI agent and an immersive XR interface (VR, AR, MR); (2) a primary pedagogical intent; and (3) reported empirical user data with a stated sample size (*N*). Non-immersive desktop simulations and purely technical architectural papers lacking user evaluation were excluded. Our multiphase screening process resulted in a final corpus of 73 articles. Data extraction parameters are summarized in Table I.

III. RESULTS

A. Overview of Included Articles

The systematic screening process yielded a final corpus of **73 high-impact articles**. The corpus is anchored by flagship venues including *ACM CHI*, *UIST*, *CSCW*, and *VRST*, alongside the core engineering and visualization tracks of *IEEE VR*, *ISMAR*, and *TVCG*.

A longitudinal analysis of the publication timeline reveals a **parabolic growth trajectory**. While XR education has been a persistent area of inquiry, the period following late 2023 marks a definitive "Generative Pivot." The significant surge in 2025 correlates with the maturation of multimodal *LLMs* and their integration into spatial computing frameworks. This trend signifies a shift from exploring "if" AI can be integrated into XR to "how" generative intelligence can fundamentally reconfigure the pedagogical affordances of immersive space.

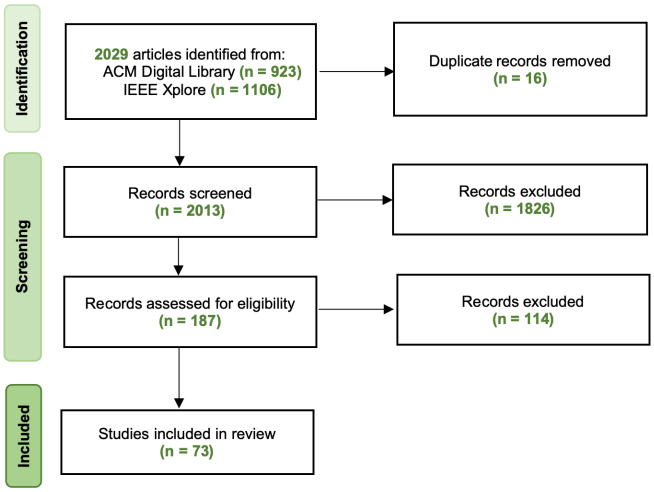


Fig. 2. Systematic review process using *PRISMA*

TABLE I
DATA EXTRACTION RUBRIC FOR THE SELECTED ARTICLES

ID	Variable	Description
DE1	Study Identifier	The unique citation identifier (Author, Source, Year).
DE2	Title	The full title of the published article.
DE3	Participant Sample (N)	The number of human participants involved in the empirical evaluation, serving as a metric for statistical power and validity.
DE4	Educational Scene	The specific learning context, such as Special Education, Training, or Cultural Heritage, classified into a taxonomy of distinct scenes.
DE5	Agent Role	The pedagogical function assumed by the Generative AI (e.g., Tutor, Peer, Co-Creator, Role-Player), defining the nature of the learner-agent interaction.
DE6	Technology Convergence	The specific *XR* modality employed (Augmented, Virtual, or Mixed Reality) and the hardware ecosystem.
DE7	Evaluation Methodology	The metrics used to assess efficacy, broadly classified into objective measures like test scores or task time, and subjective measures like NASA-TLX, self-efficacy, or presence scales.
DE8	Knowledge Domain	The academic discipline or subject matter addressed, classified into STEM and Humanities.

B. Variable Distribution

1) XR Modalities: Our analysis reveals a strategic alignment between XR technologies and distinct pedagogical objectives (Figure 3). VR dominates the corpus (64%, $n = 47$), functioning primarily as a medium for total environmental control. AR (46%, $n = 36$) serves to anchor contextual grounding, while MR (10%, $n = 7$) is emerging as a frontier for embodied co-presence. The pedagogical and psychological mechanisms driving these modal choices are synthesized further in Section.



Fig. 3. Intelligent agents across XR modalities: (a) Virtual Realityd [10]; (b) Mixed Reality [11]; (c) Augmented Reality [12].

2) Distribution of Educational Scenes: From Formal Classrooms to Ecological Contexts: The landscape of educational scenes reveals a strategic expansion from formal academic settings toward highly specialized ecological contexts. **General Education** remains the most prevalent domain, accounting for **36% (26/73)** of the corpus. These studies primarily address subject-specific challenges in K-12 and higher education, with a dominant focus on STEM disciplines where spatial visualization is paramount. For instance, Dai et al. [13] and F. -E. Boubakri et al. [14] demonstrate how immersive environments—ranging from space-themed astronomy simulations to 3D cardiac laboratories—allow students to navigate complex scientific structures. Beyond mere visualization, *GenAI* facilitates a shift toward personalized resource adaptation, as exemplified by Zhu et al. [15] through AR tools that generate customized educational content via natural language, effectively addressing the rigidity of content in traditional curricula.

Training and Vocational Education constitute the second largest category (**30%**, 22/73), focusing on high-stakes soft skills and technical proficiency. A key trend in this domain is the utilization of *GenAI* to mitigate performance anxiety, where La et al. [16] leverages generative avatars to provide a controlled environment for public speaking rehearsal. Similarly, **Cultural Heritage** (**14%**) and **Special Education** (**7%**) leverage the technology to transcend spatiotemporal and cognitive barriers. While Zhu et al. [17] utilizes persona-switching agents to deepen historical empathy, Li et al. [18] provides a critical safety net for neurodivergent learners. By simulating vocational social cues in a risk-free virtual space, these scenes offer individuals with autism a platform for social rehearsal that possesses high ecological validity without the real-world consequences of social failure.

The remaining studies explore the frontiers of **Incidental Learning** (**7%**), **Mind-Body Intervention** (**4%**), and **Psychoeducation** (**2%**). These emerging fields, such as the context-aware smart glasses developed by Cai et al. [19], suggest a future where learning is no longer confined to scheduled sessions but is instead seamlessly integrated into the user's daily interactions with the physical world. For Mind-Body Intervention, R. Kumar Kundu and K. Anuarul Hoque [20]

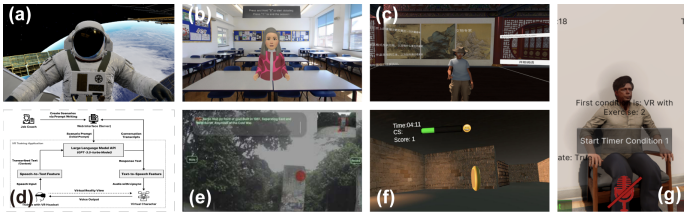


Fig. 4. Diverse educational scenes: (a) General Education [13]; (b) : Training [16]; (c) Cultural Heritage [17]; (d) Special Education [18]; (e) Incidental Learning [19]; (f) Mind-Body Intervention [20]; (g) Psycho-Education [21].

use an explainable AI and a voice-controlled dialogue engine to predict, explain, and automatically reduce cybersickness in VR. Finally, for Psycho-Education, Rings et al. [21] deploy intelligent virtual agents to guide users through psychological self-help exercises in VR, making mental health support more interactive.

3) Taxonomy of AI Roles: Mechanisms of Interaction and Scaffolding: The specific pedagogical identity assumed by a *GenAI* agent is a decisive factor in modulating learner engagement and managing cognitive load. Our analysis identifies a shift from static *NPCs* toward a nuanced spectrum of social and cognitive partnership.

The **Role-Player** emerged as the most frequent archetype (**32%**, 23/73), reflecting a significant demand for *situated social simulation*. By adopting specific personas, these agents provide a "social sandbox" for learners to practice interpersonal skills, a function particularly evident in vocational training for vulnerable populations, where agents act as workplace supervisors to facilitate the rehearsal of complex communication strategies [18]. Concurrently, the **On-demand Assistant** (**26%**) represents a reactive support model that optimizes *learner autonomy*. As demonstrated by K. Tracy and O. Spantidi [22], these agents intervene only when prompted to provide personalized guidance, thereby maintaining the user's primary cognitive flow without unnecessary distraction.

In contrast, the **Tutor** role (**23%**) assumes a proactive stance by employing structured pedagogical scaffolding to guide learners through complex knowledge domains. In virtual museum settings, for example, these agents act as expert docents that prompt reflection and interpret scientific observations in real-time [23]. A more transformative emergence is the **Co-Creator** (**11%**), which positions the AI as a creative partner. Systems like *MagicCraft* [24] empower non-technical users to shape their environments through natural language, shifting the learner's role from a passive consumer to an active producer of 3D content. Finally, the **Peer** (**5%**) and **Self-Avatar** (**3%**) roles focus on *social modeling* and embodiment, with the *ClassMeta* system [10] exemplifying how virtual classmates can model positive academic behaviors to enhance student participation through social influence. Dow et al. [25] use fully embodied AR avatars to act as remote communication partners, guiding nonspeaking autistic learners to spell on virtual letterboards.

C. The Interplay Between Agent Roles and Educational scenes

As synthesized from our cross-tabulation analysis, a significant correlation exists between specific AI agent roles and the pedagogical affordances required by different educational scenes. This alignment suggests that the deployment of Generative AI is not arbitrary but is instead strategically mapped to the specific instructional demands of the virtual environment.

1) Role-Players in Socially Complex and High-Stakes Environments: The **Role-Player** agent serves as a foundational pillar in scenarios requiring complex interpersonal navigation, most notably in **Special Education (100%)** and **Training (36.4%)**. In these contexts, the agent's core utility lies in social simulation and behavioral rehearsal. In special education for neurodivergent individuals, the agent provides a high-fidelity yet risk-free environment to practice vocational communication strategies, effectively functioning as a psychological buffer that allows learners to navigate social nuances without the anxiety of real-world failure [18], [26]. Similarly, in cultural heritage (**40%**), the Role-Player transcends historical distances by allowing participants to engage in direct dialogue with generative historical figures, thereby deepening narrative empathy through persona-based interaction [27].

2) On-demand Assistants as Scaffolds for Autonomous and Incidental Learning: The **On-demand Assistant** archetype is most prominently utilized in **Incidental Learning (60%)** and **Training (40.9%)**. This role is characterized by its reactive nature, providing scaffolding only when the learner's primary cognitive flow is interrupted by a specific knowledge gap. In incidental learning scenes, such as the use of context-aware smart glasses for everyday knowledge acquisition, the assistant remains passive until environmental stimuli trigger a need for information [19]. This strategy is equally effective in vocational training and classroom settings, where agents remain in the background until student-initiated queries occur, ensuring that pedagogical intervention remains personalized and non-intrusive [28].

3) Tutors and the Architecture of Knowledge Transmission: The **Tutor** role is the primary driver of structured knowledge acquisition, particularly dominant in **General Education (30.8%)** and **Cultural Heritage (40%)**. Unlike the reactive assistant, the Tutor assumes an omniscient perspective, employing active guidance and structured pedagogical strategies to lead learners through complex scientific or historical domains. In immersive STEM laboratories, Tutors reinforce conceptual understanding by prompting reflection and interpreting observed data in real-time, such as explaining 3D cardiac structures or microscopic morphologies [14], [23]. Furthermore, in **Psycho-education (50%)** and structured training environments, the Tutor serves as a proactive guide that transforms passive observation into an active, inquiry-based experience [21].

4) Synergies in Creative and Social Modeling: Emerging roles such as the **Co-Creator** and **Peer** provide specialized support in creative and collaborative tasks. The Co-Creator is notably active in **Mind-Body Intervention (33.3%)** and **General Education (15.4%)**, where it acts as a creative partner that empowers non-technical users to shape virtual

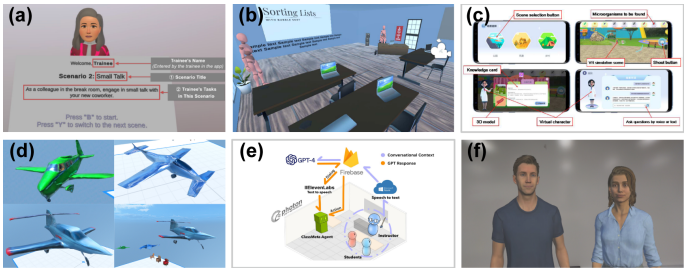


Fig. 5. Taxonomy of AI agent roles: (a) Role-Player [18]; (b) : On-Demand Assistant [22]; (c) Tutor [23]; (d) Co-Creator [24]; (e) Peer [10]; (f) Self-Avatar [25].

environments through natural language commands [29]. Concurrently, the Peer role focuses on social modeling within classroom settings, where interactive virtual classmates model positive academic behaviors to leverage social influence and reduce the communicative threshold for human students [10]. Finally, the **Self-Avatar** serves as a unique psychological tool in language education, reducing foreign language anxiety by allowing learners to interact with their own digital representation, thereby creating a sense of psychological safety [30].

5) Statistical Analysis of Role-Scenario Correlation: To further elucidate the functional alignment between AI agent roles and educational contexts, we performed a cross-tabulation analysis. As shown in Figure 6, the data reveal distinct patterns of role deployment that correspond to the specific pedagogical requirements of each scenario.

D. The Shaping of Agent Design by XR Modalities

Integrating generative artificial intelligence into XR requires a change in the architecture design of teaching agents. This section analyzes how the unique spatial insights of VR, AR, and MR determine the visual representation and behavioral design of educational artificial intelligence agents.

1) Visual Representation:

a) VR: High-Fidelity Representation within a Unified Narrative Context: In the fully immersive environment of Virtual Reality, to maintain a sense of presence, AI agents tend to employ high-fidelity 3D avatars into the narrative background to achieve educational objectives. In history education, AI agents generate authentic and switchable historical figures (such as "poets" and "archaeologists"). Furthermore, Combined with lip sync to ensure these agents speak more naturally. At the same time, combining the historical scene "Pavilion of Prince Teng" with 3D characters constructs a complete narrative field, thereby helping learners improve their learning efficiency [17].

b) MR: Malleable Representation Facilitating Co-Presence: MR emphasizes the seamless interaction between virtual environments and physical scenes, achieving image plasticity and personalized customization through GenAI, which can adapt to the needs and perception styles of different learners. For example, GenAI Agent "Kai" appears in a 3D image in the MR environment and interacts face-to-face with learners. This embodied image establishes a connection with learners and users through a sense of shared presence, thereby promoting the process of knowledge construction [31].

c) AR: The Disembodied Presentation of Situated Concepts: Compared to fully immersion of VR, AI agent systems in AR integrate 2D narrative interfaces with 3D enhanced visual effects. This approach aims to bridge abstract concepts with concrete physical environments, typically employing simplified UI designs or cartoon avatars anchored in real-world contexts. For instance, Cheng et al. [32] utilized cartoon animal agents (Oliver, Tina, and Stan) to interact with children, guiding learners to locate the real oak tree in the physical world through stylized imagery.

2) Interaction Paradigms:

a) VR: Verbal and Gaze Interaction: Verbal communication and gaze serve as the fundamental modes of interaction within authentic social scenarios. By simulating these natural mechanisms in VR, systems enhance the learner's social presence and engagement. Technical implementations typically involve a pipeline where learner speech is transcribed via APIs (e.g., OpenAI Whisper), processed by Large Language Models (e.g., GPT-3.5) for context-aware generation, and synthesized back into speech (e.g., AWS Polly). Crucially, these systems are evolving from passive responders to proactive agents; if a learner maintains prolonged gaze without verbal input, the agent can intelligently recognize this behavioral cue and initiate dialogue to sustain cognitive engagement [27].

b) MR: Physical Embodied Interaction: The AI agent architecture in MR predominantly employs a *Physical Embodied Interaction* paradigm. This functions as a cyclical feedback loop fusing the user's body, virtual information, and the physical environment. Unlike the isolated immersion of VR, this approach supports situated learning by grounding knowledge construction directly in the real world. For instance, learners may manipulate virtual objects (e.g., sliding a digital teacup) across real physical surfaces to generate context-aware feedback, thereby facilitating an intuitive understanding of physical properties and causality through tangible interaction [33].

c) AR: 2D Interface Interaction: Despite the spatial affordances of AR, the dominant interaction paradigm remains anchored in 2D Graphical User Interfaces (GUIs). Interactions are frequently mediated through overlay elements such as LLM dialog boxes, holographic keyboards, and floating menus. While many systems incorporate outdoor spatial navigation (e.g., locating campus landmarks), the core pedagogical scaffolding often relies on traditional 2D modalities. For example, [34] demonstrate this hybrid approach where students traverse physical space guided by on-screen cursors, yet verify knowledge acquisition through pop-up 2D multiple-choice windows and text-based letters from virtual characters.

E. Impact on Educational Outcomes

1) Cognitive Outcomes: The integration of AI Agents in XR has significantly enhanced cognitive outcomes, particularly within General Education, Incidental Learning, and Cultural Heritage. These enhancements are manifested through reduced cognitive load, streamlined knowledge transfer, and deepened reflective inquiry.

In General Education, K. Tracy and O. Spantidi [22] demonstrated that GPT-driven teaching assistants in VR programming

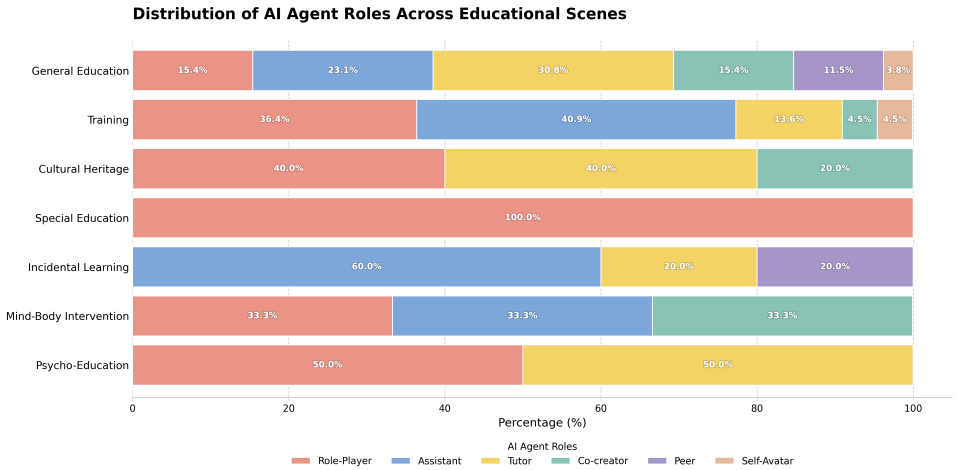


Fig. 6. Distribution of AI Agent Roles Across Educational Scenes

instruction significantly mitigated intrinsic cognitive load. This scaffolding facilitated conceptual knowledge transfer, resulting in sustained student engagement during the acquisition of complex algorithms.

Regarding Incidental Learning, the *GazePointAR* system [35] illustrates how multimodal input (gaze tracking combined with voice) reduces the cognitive threshold for information retrieval. By instantly analyzing environmental queries (e.g., "What is this?"), the agent enables intuitive, context-aware knowledge acquisition in mobile scenarios.

In Cultural Heritage, AI Agents promote deep reflection by constructing multi-perspective narratives. Lau et al. [36] found that personalized, agent-driven narratives in oral tradition contexts stimulated deeper self-reflection, significantly enhancing comprehension of unfamiliar cultural knowledge compared to static descriptions.

2) Skill Outcomes: AI Agents in XR exert a transformative impact on Skill Outcomes, most notably in Training and Special Education, by providing personalized feedback and high-fidelity situational simulation.

In Training contexts, such as sports, agents improve both performance metrics and psychological adaptability. For instance, in Sabre Fencing, AI agents provide kinematic analysis that augments human coaching, offering detailed insights into technical movements [37].

In Special Education, particularly for individuals with Autism Spectrum Disorder (ASD), these agents function as critical social scaffolds. . Li et al. [26] report that VR interview simulators allow learners to practice vocational communication skills in a distraction-free, risk-free environment. This rehearsal builds essential self-efficacy before learners transition to real-world social interactions.

3) Affective Outcomes: In Psycho-education and Mind-Body Interventions, AI Agents are instrumental in alleviating stress and fostering emotional connection.

In Psycho-education, agents establish strong therapeutic alliances, preventing the isolation often felt during self-guided psychological exercises. Notably, Spatially Aware LLM Agents can integrate physical environmental elements (e.g., a toy in the room) into the dialogue. This acute environmental awareness enhances Affective-based Trust and encourages self-disclosure, fostering a sense of calm and safety [38].

In Mind-Body Interventions, the convergence of XR and AI constructs a psychological safety net for emotional regulation. Fang et al. [39] utilized agents to guide users through stress relief techniques (e.g., "box breathing") during simulated high-pressure scenarios like public speaking. This real-time intervention effectively reduces anticipatory anxiety, enhancing confidence and emotional resilience.

F. Evaluation Methods and Metrics

Efficacy evaluation across the corpus predominantly employs a mixed-methods approach. **Subjective measures** (96% of studies) are the standard, utilizing semi-structured interviews alongside validated usability instruments like the System Usability Scale (SUS) and NASA-TLX to quantify cognitive workload. **Objective performance metrics** (74%) complement these through knowledge retention quizzes and task efficiency completion rates. Notably, an emerging subset (21%) integrates **physiological and behavioral signals**—such as HMD-integrated eye-tracking for visual attention and fNIRS for hemodynamic responses—signaling a shift toward granular, objective biometric assessment.

IV. DISCUSSION

A. The Reshaping of AI Agent Role by XR Modalities

1) The Dominance of Environmental Control: VR as a Generative Sandbox: Our analysis establishes Virtual Reality (VR) as the hegemonic modality, accounting for 64% of the corpus. This dominance is not merely a reflection of hardware availability but indicates a pedagogical preference for *Total Environmental Control*. In domains such as microbiology and astronomy, VR is employed to eliminate external noise, functioning as a "generative sandbox" where learner agency is unconstrained by physical laws. Unlike traditional simulations that rely on static decision trees, GenAI-enhanced VR transforms the environment into a responsive entity. For

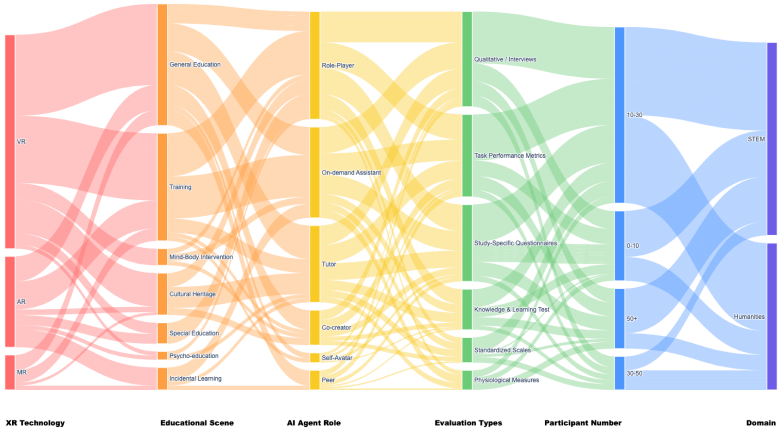


Fig. 7. Sankey Diagram, a visualization showing the distribution of characteristics across all dimensions in the reviewed literature. Different stages (nodes) are color-coded to represent distinct dimensions: ■ **XR Technology**, indicating the types of extended reality used (e.g., VR, AR, MR); ■ **Educational Scene**, categorizing specific learning contexts (e.g., General Education, Training); ■ **AI Agent Role**, highlighting the specific functions AI agents serve (e.g., Role-Player, Tutor, On-demand Assistant); ■ **Evaluation Types**, illustrating assessment methods such as qualitative interviews or task performance metrics; ■ **Participant Number**, showcasing the size of participant groups in the studies; and ■ **Domain**, representing academic application areas such as STEM or Humanities. The width of the flows (links) is proportional to the total number of reviewed articles (N = **73**), reflecting the prominence and interconnectedness of these research dimensions.

instance, microscopic biological models no longer just display animations but respond bidirectionally to learner inquiries, effectively transitioning from "viewing" a simulation to "inhabiting" a living system [23]. This suggests that the primary affordance of GenAI in VR is the creation of a closed-loop, high-fidelity narrative field that sustains the "illusion of non-mediation" [10], [40].

2) Ontological Security in Augmented Reality (AR): In contrast, Augmented Reality (AR) (46% of studies) operates on a paradigm of *Contextual Grounding*. Our synthesis suggests that AR is preferred not for immersion, but for *Ontological Security*—the preservation of the learner's connection to the physical world. By anchoring generative intelligence to real-world objects, AR reduces the "translation gap" between abstract concepts and concrete application. This is particularly critical in special education, where the complete isolation of VR can induce sensory overload. Studies demonstrate that autistic learners interact more effectively with virtual agents when they remain anchored in a familiar physical space, using the predictable reality as a psychological safety net while engaging with the unpredictable AI [11].

3) Mixed Reality (MR): The Frontier of Hybrid Focus: Mixed Reality (MR), though nascent (10%), introduces the capability of *Spatial Filtering*. Unlike AR, which adds to reality, MR can selectively subtract from it. Research indicates that MR agents can dynamically filter physical distractions to create "hybrid focus zones," balancing the need for physical tool manipulation with the cognitive depth required for reflective tasks [12]. This positions MR as the ideal modality for "Embodied Co-learning," where the agent physically occupies the user's workspace, fostering a sense of co-presence that pure VR or AR cannot achieve [31], [41].

B. The Pedagogical Spectrum of Agent Role Design

1) Shifting Power Dynamics: From Authority to Partnership: The integration of GenAI is fundamentally reshaping the power dynamics of the classroom. While the **Omniscient Tutor** (23%) remains a staple for optimizing cognitive load through direct instruction, there is a marked shift towards the **Collaborative Peer** (5%) and **Role-Player** (32%). This transition represents a move from "instruction" to "social modeling." By simulating a peer who is also "learning" or a persona within a specific social context, these agents effectively reduce the authority gradient [10], [42]. This encourages learners to engage in critical thinking and error modeling without the fear of judgment associated with expert tutors. This "Learning Companion" archetype leverages *Social Presence* to maintain motivation, transforming the solitary act of studying into a communal experience [43].

2) From Consumption to Creation: The Co-Creator Archetype: A distinct "Generative Pivot" is observed in the rise of the **Co-Creator** role (11%). Traditional educational technology often positions the learner as a passive consumer of content. However, GenAI agents in this role act as "force multipliers" for creative intent. By interpreting natural language commands to generate complex 3D assets, these agents lower the technical barrier to entry for spatial design [15], [24]. This shifts the pedagogical focus from information retrieval to *Generative Inquiry*, where learning occurs through the iterative process of describing, visualizing, and refining ideas in 3D space [29], [44].

3) The Agent as "Self": Externalizing the Ego: A transformative, albeit less common, application is the **Self-Avatar** (3%). In high-anxiety domains like foreign language learning, interacting with a digital representation of oneself creates a unique psychological distance. This "externalization of the ego" allows learners to view their performance objectively and practice socially risky interactions with a sense of psychological safety [25], [30]. This suggests a new theoretical avenue where GenAI serves not just as an interlocutor, but as a mirror for self-reflection.

C. Multidimensional Mechanisms of Learning Efficacy

1) Cognitive Efficiency via Adaptive Scaffolding: The primary cognitive mechanism identified is *Adaptive Scaffolding*. Unlike rigid rule-based systems, GenAI agents function as probabilistic reasoning engines that construct knowledge incrementally. This capability allows them to dynamically modulate the complexity of information based on the learner's immediate cognitive load. For instance, in programming education, agents that handle low-level syntax retrieval significantly reduce intrinsic cognitive load, freeing up mental resources for high-level algorithmic logic [45], [46]. Furthermore, multimodal agents (combining gaze and voice) reduce the "search cost" of information in mobile contexts, making knowledge acquisition intuitive and incidental [19], [35].

2) Deepening Trust through Spatial Awareness: We argue that *Social Presence* in XR is qualitatively different when mediated by GenAI. It transitions from "reading text" to "feeling accompanied." A critical factor here is *Spatial Awareness*. Agents that can perceive and reference objects in the learner's physical room (e.g., a toy or furniture) bridge the gap between the virtual and physical, establishing "Affective-based Trust" [38], [47]. This acute environmental awareness encourages greater self-disclosure in therapeutic settings, as the agent is perceived as sharing the learner's physical reality rather than just simulating a conversation.

3) Embodied Action and Skill Mirroring: Finally, the combination of XR and AI facilitates *Embodied Action*. In psychomotor domains, agents do not just explain movements; they mirror them. In fencing training, for example, agents provide kinematic data that helps learners internalize complex physical forms [37]. Similarly, in vocational training for neurodivergent individuals, agents provide a "social sandbox" for behavioral rehearsal. This allows learners to practice workplace communication strategies in a safe, simulated environment before transferring these skills to high-stakes real-world interactions [26], [48].

D. Practical Applications and Design Guidelines

1) Strategy for Modality Selection: The selection of an XR modality should be driven by the specific psychological and environmental requirements of the learning task. VR is recommended when the pedagogical goal requires total environmental control, functioning as a "generative sandbox" where high-stakes or inaccessible scenarios can be simulated without external noise. In contrast, AR should be prioritized for tasks requiring "ontological security," as it preserves the learner's connection to the physical world, which is critical for mitigating sensory overload and environmental anxiety. MR serves as the frontier for "embodied co-learning," offering spatial filtering capabilities that create "hybrid focus zones" ideal for tasks requiring the seamless coexistence of digital intelligence and physical tool manipulation.

2) Principles for Role Design: The pedagogical identity of a generative AI agent must be strategically mapped to the instructional demands of the environment. Role-Players should be deployed in socially complex environments to provide a risk-free "social sandbox" for behavioral rehearsal and empathy building. On-demand Assistants are most effective for autonomous and incidental learning, adopting a reactive stance to provide scaffolding only when the learner's cognitive flow is interrupted. For structured knowledge transmission, the Tutor role is preferred, employing proactive guidance and structured pedagogical strategies to lead learners through complex domains. Finally, the Co-Creator archetype acts as a "force multiplier" for creative intent, shifting the learner from a passive consumer to an active producer through natural language-driven content generation.

3) Orchestrating Multiple Roles and Modalities: Effective system design requires orchestrating multiple roles and modalities to balance immersion, accuracy, and engagement. Designers should implement context-aware role-switching, allowing agents to transition between archetypes to reduce authority gradients and foster critical thinking. To address the risk of generative hallucinations, especially in high-stakes STEM subjects, systems should integrate hybrid verification strategies such as Retrieval-Augmented Generation (RAG) or 2D GUI overlays that cross-verify agent outputs against established standards. Lastly, the interface must bridge the "prompting barrier" by moving toward intent-based interaction, leveraging XR's multimodal affordances—such as gaze tracking and hand gestures—to infer learner needs implicitly without requiring complex linguistic input.

V. CHALLENGES AND FUTURE DIRECTIONS

A. Future Direction: Overcoming Technical Bottlenecks

1) Latency and Synchronization in Multi-modal Interaction: The current modular processing architecture creates a significant technical bottleneck that hinders seamless immersion. As identified in our analysis, the standard workflow typically relies on a cascaded series of discrete steps: transcribing speech into text, processing that text through a Large Language Model for inference, synthesizing the response back into speech, and finally generating corresponding lip movements [48], [49]. This fragmented, multi-stage process introduces cumulative delays that disrupt the natural rhythm and turn-taking dynamics of conversation, thereby shattering the "illusion of non-mediation" [28], [50]. Future research must prioritize the transition toward unified, end-to-end multimodal models that integrate perception and generation layers. By minimizing inference latency and ensuring the temporal synchronization of verbal and non-verbal cues, these advanced systems can sustain the visceral sense of co-presence necessary for effective social scaffolding.

2) AI Hallucinations and Pedagogical Accuracy: The shift from deterministic, rule-based Intelligent Tutoring Systems (ITS) to probabilistic generative models introduces the inherent risk of "hallucinations"—plausible but factually incorrect outputs [33], [44]. Unlike traditional systems where instructional designers hard-code every response [14], [51], GenAI agents operate as stochastic reasoning engines. In high-stakes educational domains such as STEM, this poses a significant pedagogical risk. For instance, in 3D cardiac anatomy simulations [14] or contextualized mathematical problem generation

[45], a "plausible but wrong" explanation regarding physiological structure or algorithmic logic could lead to ingrained misconceptions that are difficult to correct.

Similarly, in Cultural Heritage and Humanities education, the stochastic nature of LLMs conflicts with the requirement for historical and cultural fidelity. When agents simulate historical figures or engage in oral storytelling [36], [52], unverified generative outputs risk engaging in historical revisionism or cultural misrepresentation, thereby undermining the learning objective of empathy and accuracy.

To mitigate these risks, future systems require the integration of robust "Retrieval-Augmented Generation" (RAG) frameworks and expert-in-the-loop verification mechanisms. Emerging research demonstrates the efficacy of "Tool-Augmented" architectures, where the autonomous agent is grounded by external, verified databases to ensure accuracy while retaining the flexibility of natural language [15]. Furthermore, hybrid verification strategies, such as those employing pop-up multiple-choice windows to cross-verify generative narratives against established curriculum standards, are essential to ensure that the content generated by autonomous agents remains scientifically rigorous and historically accurate [53].

B. Interaction Design Gap: From Intelligence to Embodiment

1) Non-verbal Expressiveness and Emotional Fidelity: A significant dissonance exists between the high linguistic intelligence of modern agents and their "embodied stupidity." While LLMs excel at generating context-aware text, a gap persists in *Embodied Expressiveness*. Current agents often display repetitive, generic, or context-mismatched gestures that fail to convey the emotional nuance of human interaction [26]. To achieve true "Social Presence," future work must move beyond pre-scripted animations. This involves integrating affective computing with procedural animation and Inverse Kinematics (IK) to enable agents to display micro-expressions, coherent gaze patterns, and culturally appropriate proxemics that align dynamically with the emotional tone of the conversation [54].

2) Bridging the "Prompting Barrier" for Non-Technical Users: The efficacy of GenAI agents is currently gated by the user's ability to craft precise and effective prompts. This creates a "Prompting Barrier" or digital divide for non-technical learners, forcing them to act as "prompt engineers" rather than students [26]. Future interface designs must move towards "Intent-Based Interaction." Instead of relying solely on explicit text or voice commands, systems should leverage the multimodal affordances of XR—such as gaze tracking, hand gestures, and object manipulation—to infer learner intent implicitly. By analyzing *where* a learner is looking or *what* they are holding, the agent can provide proactive assistance without requiring complex linguistic input [35].

C. Pedagogical Validity: The Need for Longitudinal Evidence

Despite the promising theoretical potential to address Bloom's "2 Sigma Problem" [1], there is a scarcity of empirical evidence validating the long-term efficacy of GenAI in XR. The majority of existing studies focus on short-term usability, immediate engagement, or self-reported satisfaction [10], [23], [30], [55]. To establish GenAI-XR as a legitimate pedagogical paradigm, the field requires rigorous longitudinal studies. These inquiries must move beyond simple knowledge retention to measure the *Transfer of Learning*—specifically, the extent to which skills acquired in the generative simulation (e.g., conflict resolution, surgical precision) are successfully applied in real-world performance contexts over time.

D. Ethical and Social Implications

As agents become more anthropomorphic and emotionally responsive, deep ethical concerns regarding "Emotional Dependency" and the nature of the human-agent relationship arise. In sensitive contexts such as Psycho-education and Special Education, where agents establish a strong "Therapeutic Alliance" [38], there is a risk of users blurring the line between a pedagogical tool and a social companion. Furthermore, the extensive collection of biometric data (e.g., eye-tracking, physiological signals) required for adaptive scaffolding raises significant privacy concerns. Future research must investigate the psychological impact of forming social bonds with AI entities and establish ethical guidelines to prevent over-reliance, ensuring that these systems augment rather than replace human social connection.

VI. Conclusion

This systematic review maps the paradigm shift toward dynamic, "Generative XR" pedagogical ecosystems. By synthesizing 73 articles, we establish a functional taxonomy of embodied agents (Tutors, Peers, Role-Players, and Co-creators) to ground human-AI interaction in immersive spaces. Our analysis highlights how XR modalities shape these interactions: VR functions as a high-fidelity "generative sandbox," whereas AR and MR facilitate ontological security and embodied co-presence. Together, these systems yield significant benefits, including adaptive cognitive scaffolding, enhanced social self-efficacy, and democratized content creation. To realize the full potential of situated learning, however, future work must address technical bottlenecks—such as multimodal latency and AI hallucinations—through hybrid verification mechanisms. Advancing from explicit prompting to implicit, intent-based interactions will be critical to minimizing the "interaction gap". Supported by future longitudinal efficacy studies, the convergence of Generative AI and XR promises to fundamentally redefine global educational infrastructure, making intelligent embodiment a ubiquitous learning component.

Supplemental Material

The supplementary material associated with this article includes a detailed data extraction list mapping the 73 included publications to their respective educational scenes, agent roles, and XR modalities. This material can be found in the online version of this article.

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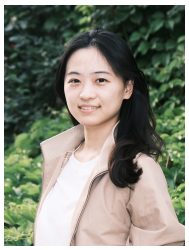
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